

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

project_overview_plots.py

These are the summary figures, meant to give a quick overall picture of what data exists and what was found, without having to open any of the detailed outputs.

- **project_sample_design.png:** two panels. The first is a bar chart of how many fish there are in each age and genotype group. The second is a grid showing which combinations exist and which are missing. The gaps are the important part, because there are only mutants at 3 and 18 months and only wild-types at 11 and 37 months, and that is what decides which questions can be asked at all. It is why the interaction model uses 3 and 18 months and why the ageing question uses only wild-type fish.
- **project_tf_counts.png:** the number of significant transcription factors for each comparison, split into activated and repressed. The 3-month genotype bar is empty, the 18-month genotype bar has 71 TFs, and the ageing bar has 37. The empty bar is the most informative thing on the chart. One point worth stating clearly is that TF activity is always a property of a comparison between two groups, never of a single group on its own, so there is no such thing as the number of TFs in the mutant group or at 18 months by itself. That is why the plot is organised by comparison.
- **project_results_overview.png:** a four panel dashboard. The first panel is the number of significant genes in each comparison, showing 9 at 3 months against 5,541 at 18 months. The second is the interaction patterns, where age-dependent onset dominates and constitutive is zero. The third is how many genes are affected by age, genotype, both or neither. The fourth is the GRN target enrichment. Together they read as a summary of the entire project: the mutation is silent in young fish, disruptive in old fish, and the transcription factors driving it are the ones controlling the age-dependent gene programme.